FUNCTIONAL TEST CASES – GLOF MONITORING SYSTEM

1. Data Acquisition

Feature	Test Case	Steps to execute test case	Expected Output	Actual Output	Status	More Information
Data Acquisition	Load Satellite Data	Open system → Select dataset → Load Sentinel-2 data	Data should load successfully	Data loads correctly	Pass	Ensure no missing bands
Data Acquisition	Historical Data Load	Load Landsat-5 dataset	Historical data (1988–2023) should be available	Data loaded	Pass	Verify temporal consistency

2. DEM Processing

Feature	Test Case	Steps to execute test case	Expected Output	Actual Output	Status	More Information
DEM Processing	Sink Filling	Input DEM → Run sink filling	All artificial sinks should be removed	Sink filling executed	Pass	Ensure no elevation gaps
DEM Processing	Flow Direction	Run flow direction algorithm	Each pixel should have valid flow direction	Correct flow directions generated	Pass	Uses D8 logic
DEM Processing	Flow Accumulation	Run accumulation	High accumulation areas should represent rivers	Output matches expected drainage	Pass	Check river paths
DEM Processing	Watershed Extraction	Select pour point → Run watershed	Micro-watersheds should be generated	413 MWs generated	Pass	Verify boundary accuracy

3. Spectral Index Computation

Feature	Test Case	Steps to execute test case	Expected Output	Actual Output	Status	More Information
Index Computation	NDWI Calculation	Input bands → Compute NDWI	Water bodies should be highlighted	Correct NDWI values	Pass	Range: 0.1–0.5 for water
Index Computation	NDSI Calculation	Input bands → Compute NDSI	Snow/ice should be detected	Correct classification	Pass	Threshold ≥ 0.4

4. Machine Learning Classification

Feature	Test Case	Steps to execute test case	Expected Output	Actual Output	Status	More Information
ML Classification	Decision Tree Classification	Input spectral data → Run DT model	Pixels classified into Glacier, Lake, Non-glacier	Correct segmentation	Pass	No overlap
ML Classification	Accuracy Validation	Compare output with known data	High accuracy classification	Matches expected	Pass	Reduces false positives

5. Morphometric Computation

Feature	Test Case	Steps to execute test case	Expected Output	Actual Output	Status	More Information
Morphometric Analysis	AAR Calculation	Compute accumulation area / glacier area	Valid AAR values generated	Correct AAR	Pass	Indicates glacier health
Morphometric Analysis	Snout Index	Compute elevation difference	SI values should reflect slope	Correct SI	Pass	High SI = high risk
Morphometric Analysis	CCI Calculation	Compute catchment compactness	Circular basins identified	Correct CCI	Pass	High CCI = faster runoff

6. GLOF Index Generation

Feature	Test Case	Steps to execute test case	Expected Output	Actual Output	Status	More Information
GLOF Index	Risk Calculation	Input all parameters → Run model	Risk levels assigned (5 categories)	Correct classification	Pass	Based on DT model
GLOF Index	Hazard Mapping	Generate map	Spatial hazard zones displayed	Map generated	Pass	Verify zones

7. Warning System

Feature	Test Case	Steps to execute test case	Expected Output	Actual Output	Status	More Information
Warning System	Warning Generation	Run final pipeline	Warning levels generated for all MWs	Output generated	Pass	413 MWs covered
Warning System	High Risk Detection	Filter high-risk lakes	High-risk lakes identified	2 lakes detected	Pass	Matches dataset
Warning System	Final Report	Export results	Report generated (.xlsx)	File created	Pass	Check format